

Object Oriented Programming

Dr. Imran Ashraf

Office: 02

`imran.ashraf[at]hitecuni[dot]edu[dot]pk`

Review

- Functional Programming
- Structured Programming
- Imperative Programming
- Object Oriented Programming

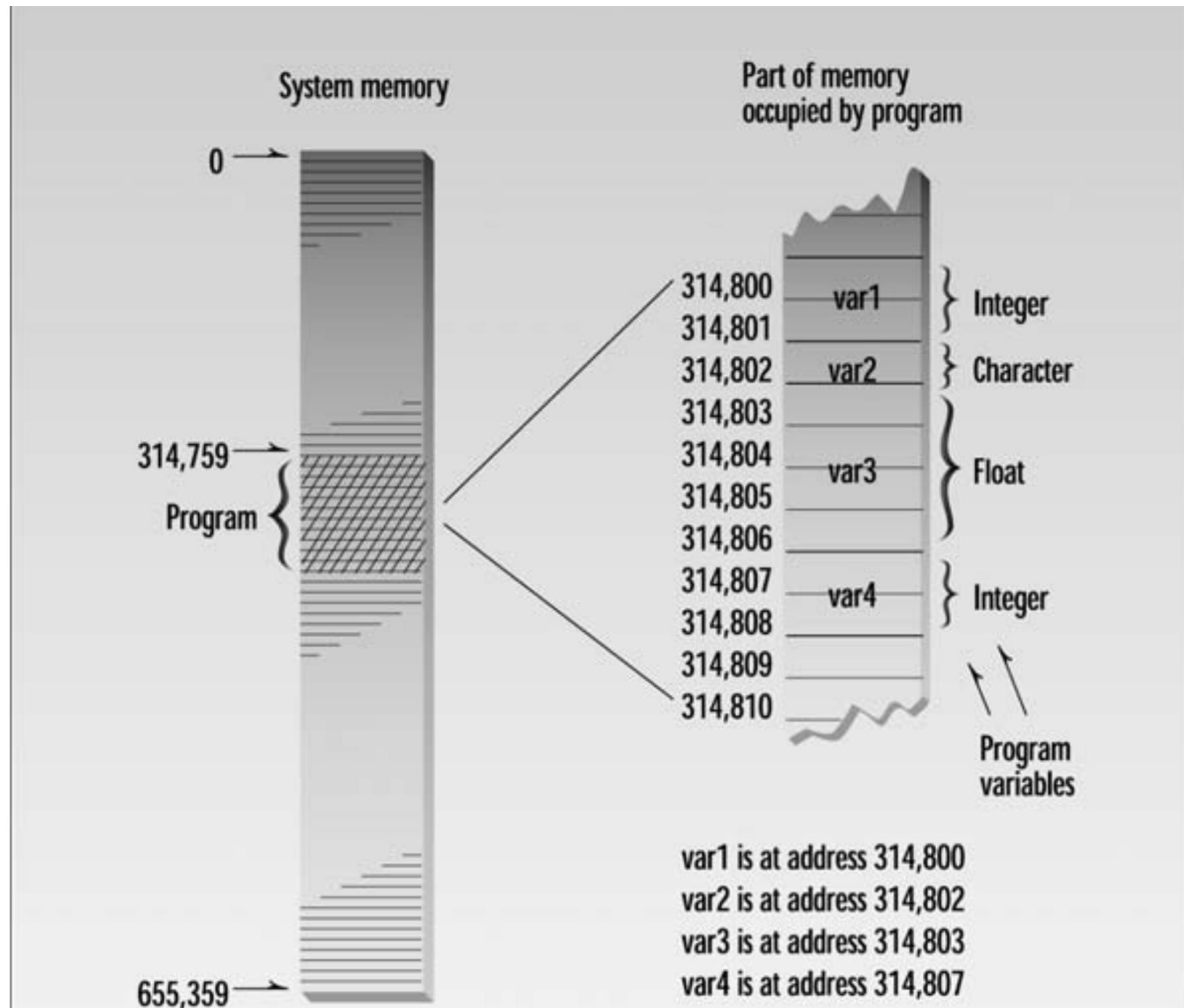
Today's Lecture

- Pointers
- References
- Functions
 - Passing arguments to functions
 - Function Overloading

Pointers

- An important concept in C/C++
 - Variable name
 - Variable address

Programmer's view on memory



Working with pointers

- Address operator
- Pointer variable
- Value operator

Address operator

- How can we print address of something -
Address operator &
- **cout << &var1 << endl**
 - **<< &var2 << endl**
 - **<< &var3 << endl;**

Pointer Variable

- `int * ptr1;`
- `float * ptr2;`
- `unsigned char * ptr3;`

Value operator

- `cout << *ptr1;`
- `int b = *ptr1;`

Questions/Discussion

- Why pointers are powerful?
- Why pointers can be dangerous?
- Why pointers should always have value?

Use of Pointers

- Accessing array elements
- Passing arguments to a function when the function needs to modify the original argument
- Passing arrays and strings to functions
- Obtaining memory from the system
- Creating data structures such as linked lists

Use of pointers: Array access

- How elements of an array can be accessed by using pointers?

Use of pointers: Structure access

- How elements of a structure can be accessed by using pointers?

Passing arguments to functions

- By value
- By reference
- By pointers

Memory allocation

- Static allocation
- Dynamic allocation
 - new
 - delete

Summary

- Pointers are powerful concepts
- Utilization of pointers
 - Address operator
 - Pointer variables
 - Value operator
- Use of pointers

Next Lecture

- Classes and Objects